1. MODEL'S MAIN GEOMETRY AND VARIABLES DEFINITIONS

Figure 1 shows the model's main geometry and varaible definitions. It is a simple frame, composed of 4 blocks (27.3 × 50.3 × 25.5 mm for height length and width) of Tecnil polymer with masses of m_i=122.2 grams, measure with a scale model XXX). The blocks are fixed by steel rulers using 4 screws in each side. The rulers have an estimated Young modulus of E =2.1×10^{11} N/m^2. The thickness is t = 0.45 mm, the width is w = 25.5 mm measured with a digital caliper model CFC Eletronic Digial Capiler (150mm stroke) with resolution of 0.1 mm and accuracy of ±0.2 mm. The used data acquisition software used was the Agilent Vee 4.0 connected with a data acquisition board USB1208–FS by Measurement Computing. Four accelerometers (model ADXL203 from Analog Devices) were attached at the top of each block using bee wax for measuring lateral vibrations. The acceleration and frequency range of accelerometer are ±1,7gravits and 0-2 kHz, respectively. The sampling rate was set to f_s = 600 Hz. The samples were acquired fixing the model to a rigid wood table using clamps and the measurements happened with different operators, in different days and moments of the day, always disassembling and assembling the setup for measurements in blocks of 10 measurements, each time. Figure 2 shows the setup for the measurements.

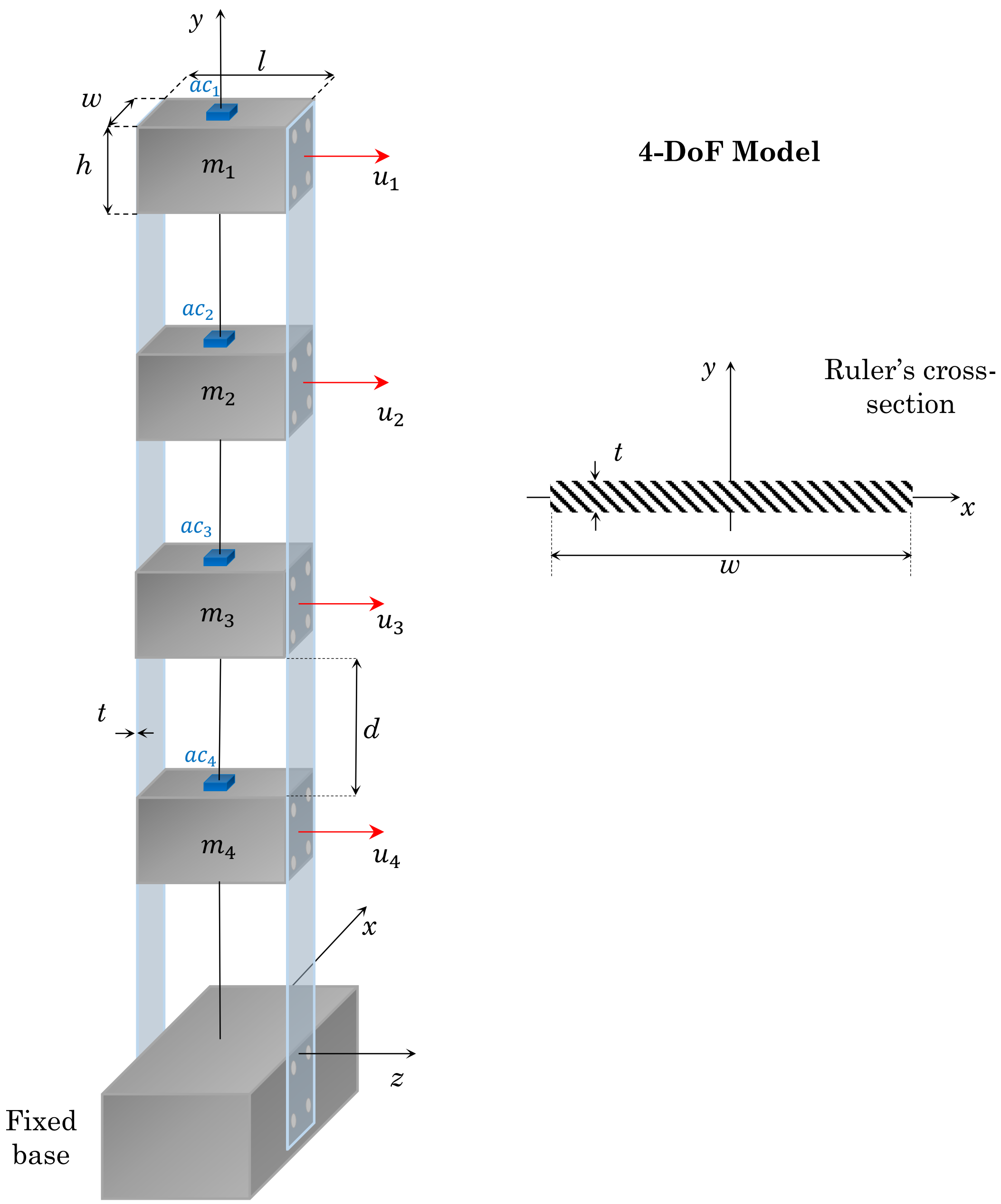

Figure 1 – Variable definitions and main parameters of the model.

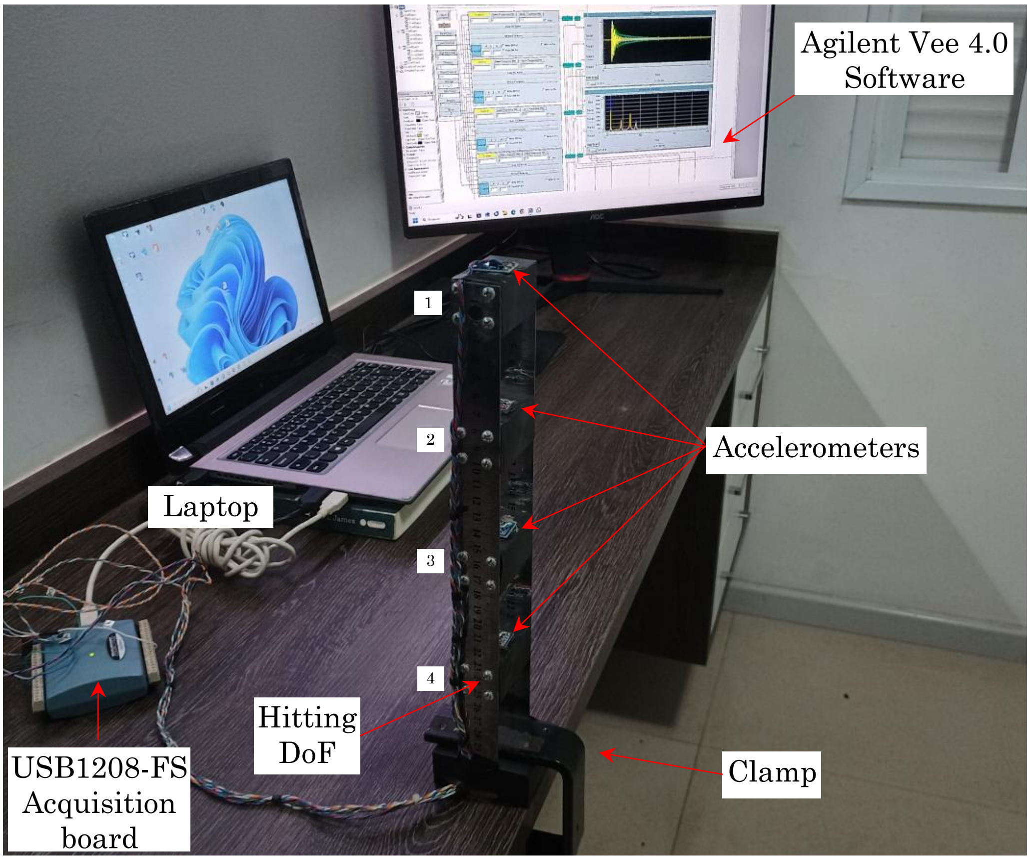

Figure 2 – Setup for the measurements.

2. EQUIPMENT AND MATERIALS USED IN MEASUREMENTS